



Module 21 Day 4: Evidence Acquisition and Reports

Class Overview

Today we will conclude our introduction to digital forensics. You will continue the investigation of Peter's iPhone by examining emails and text messages and produce a preliminary log and report on the evidence found.

Class Objectives

- Use Autopsy to view and gather evidence from Peter's emails.
- Use data exports to analyze email messages offline.
- Use data exports to analyze SMS messages offline.
- Use Autopsy to extract GPS data and identify WiFi locations.
- Prepare a preliminary report using evidence worksheets.

Lab Environment

Lab Details

In this module, you will continue to use the Forensics lab environment.

User `kali` Password `kali`

Slideshow

- Slides for today's lesson can be found on Google Slides: [21.4 Slides](#).

01. Welcome and Overview

Welcome to the third day of the digital forensics module. Let's review what we covered in the last class:

- Identifying methods used in smartphone forensics
- Examining databases and file structures of an iPhone directory image
- Locating and documenting evidence of ownership for an iPhone image
- Using Autopsy to view and tag evidence of an iPhone image
- Extracting data for use in other applications (logs, text, pictures, video, audio)

We will begin class with a warm-up activity.

- As a forensic investigator, you must understand how to use tools such as Autopsy and how to export data so other team members can perform offline analysis.
- .emlx is a file extension called Mail Message that's used to store email messages.
- EMLX files are often referred to as Apple Mail files because they are created with Apple's mail program to store plain text files for a single message.

02. Activity: Email Export

- In this activity, you will continue your role as a digital forensic investigator.
- Your task is to export the email database for offline analysis using examination tools outside of Autopsy.

- [Activity File: Email Export](#)

03. Activity Review: Email Export Activity

04. Instructor Do: Peter's Email Evidence

We exported Peter's emails to the `Export` directory during the warm-up activity. Now, we'll use offline analysis to uncover details about the plot of the crime and possibly identify the co-conspirators and their roles.

Importance of Emails

We can extract critical information from emails, such as:

- Sender's email address
- Sender's IP address
- Internet service provider
- User client (email app)
- Location information

All this information plays an important role in conducting a digital forensics investigation.

Examine Emails in Kali Linux

We will demonstrate how to display email messages in the `Mail/MessageData` folder.

- These emails were exported to the `Mail/MessageData` folder in Kali Linux during the warm-up activity.
 - *Note: There may be a small variation in the name of your "mail" folder.*

Open a terminal window and navigate into the directory that contains the `MessageData` directory.

- Run `ls -l` to see the full contents of the `Messages` folder.

Within each of these directories contains EMLX files.

```
(kali㉿kali)-[~/.../Pure-Gold-Credit-Union/Export/Mail/MessageData]
└─$ ls -l
total 88
drwxr-xr-x 2 kali kali 4096 May 13 16:44 1
drwxr-xr-x 2 kali kali 4096 May 13 16:44 10
drwxr-xr-x 2 kali kali 4096 May 13 16:44 11
drwxr-xr-x 2 kali kali 4096 May 13 16:44 12
drwxr-xr-x 2 kali kali 4096 May 13 16:44 13
drwxr-xr-x 2 kali kali 4096 May 13 16:44 14
drwxr-xr-x 2 kali kali 4096 May 13 16:44 15
drwxr-xr-x 2 kali kali 4096 May 13 16:44 16
drwxr-xr-x 2 kali kali 4096 May 13 16:44 17
drwxr-xr-x 2 kali kali 4096 May 13 16:44 18
drwxr-xr-x 2 kali kali 4096 May 13 16:44 19
drwxr-xr-x 2 kali kali 4096 May 13 16:44 2
drwxr-xr-x 2 kali kali 4096 May 13 16:44 20
drwxr-xr-x 2 kali kali 4096 May 13 16:44 21
drwxr-xr-x 2 kali kali 4096 May 13 16:44 22
drwxr-xr-x 2 kali kali 4096 May 13 16:44 3
drwxr-xr-x 2 kali kali 4096 May 13 16:44 4
drwxr-xr-x 2 kali kali 4096 May 13 16:44 5
drwxr-xr-x 2 kali kali 4096 May 13 16:44 6
drwxr-xr-x 2 kali kali 4096 May 13 16:44 7
drwxr-xr-x 2 kali kali 4096 May 13 16:44 8
drwxr-xr-x 2 kali kali 4096 May 13 16:44 9
```

Search within these directories for full.emlx files.

Search for items of interest

In the next activity, you will continue your investigation by using preview commands to view the contents of each EMLX file, as shown below.

```

(kali@kali)~[~/Export/Mail/MessageData/10]
$ more full.emlx
To: rosielloyd071292@icloud.com
From: Peter Barnes <peterbarnes12792@icloud.com>
Subject: Dinner
Date: Thu, 12 Oct 2023 00:47:25 +0000 (UTC)
X-Client-IP: 10.182.149.140
X-Mailer: iCloud MailClientcurrent MailServer2322B155.10000-Famine2322-0-e29a334e875d
Message-id: <0bf02314-7420-4014-0f4-c16f0e96844d@me.com>
Content-Type: multipart/alternative; boundary=Apple-Webmail-42--7cf28b23-0202-4563-8c70-36f17308df19
MIME-Version: 1.0

--Apple-Webmail-42--7cf28b23-0202-4563-8c70-36f17308df19
Content-Transfer-Encoding: quoted-printable
Content-Type: text/plain;
    charset=utf-8;
    format=flowed

Hey Rosie, great hanging out with you this weekend! Always good to hang out
and talk about non work things.Peter Barnes
--Apple-Webmail-42--7cf28b23-0202-4563-8c70-36f17308df19
Content-Type: multipart/related;
    type="text/html";
    boundary=Apple-Webmail-86--7cf28b23-0202-4563-8c70-36f17308df19

--Apple-Webmail-86--7cf28b23-0202-4563-8c70-36f17308df19
Content-Transfer-Encoding: quoted-printable
Content-Type: text/html;
    charset=utf-8;

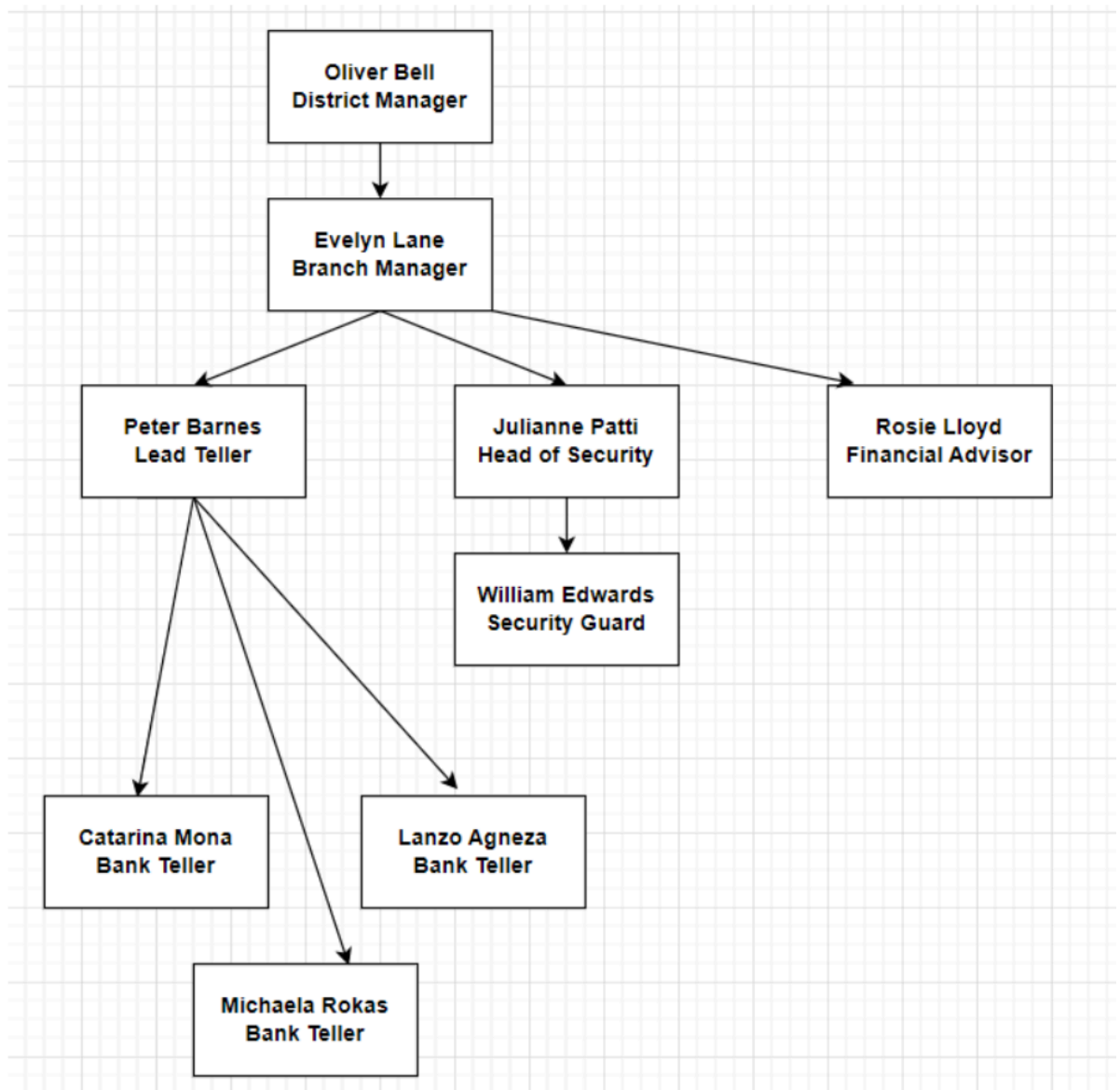
<html><body><div>Hey Rosie, great hanging out with you this weekend! Always good to ha=
ng out and talk about non work things.<br></div><div><br></div><div>Peter =
Barnes</div></body></html>
--Apple-Webmail-86--7cf28b23-0202-4563-8c70-36f17308df19--
--Apple-Webmail-42--7cf28b23-0202-4563-8c70-36f17308df19--

```

- You will also look at any email attachments and tag items of interest in Autopsy.

The following diagram provides a high-level overview of the threat actor's involvement in the Pure Gold Credit Union case.

- Use the following as a roadmap to tie Peter's associates to each incident.



- The [Correspondence Evidence Worksheet](#) is used to document evidence from the communications (i.e., phone calls, emails, text messages) found on Peter's phone. You should complete this as a group.
- You will fill out the following information:
 - Artifact numbers to help organize records
 - Timestamp (time email was sent or received)
 - Header information
 - These should include the names of the individuals involved, their email addresses, and the email subject line.
 - Key information
 - Summary of the email contents

- Evidence location
 - Source of the data (For example, email, SMS, phone call, etc.)

Master Timeline of NGDC				
Artifact #	Timestamp	Header Information	Key Information	Evidence Location
1	10/12/23 00:47:25	To; rosielloyd071282@icloud.com From: peterbarnes12792@icloud.com Subject: Dinner	Peter emailed Rosie to tell her it was good to hang out and talk about “non work” things	MessageData/10

05. Activity: Peter's Email Evidence

- In this activity, you will use Autopsy to view Peter's email correspondence and generate a list of contacts and their email addresses.
- You will be assigned to work in a small group as part of a digital forensics team.
- You will create custom tags for the evidence in Autopsy.
- You will also document your findings using the Correspondence Evidence Worksheet.
- Remember that these emails were exported to the Mail/MessageData directory during the warm-up activity. This is where you can perform offline analysis.
- [Activity File: Peter's Email Evidence](#)

06. Break

07. Activity Review: Peter's Email Evidence Activity

08. Peter's SMS Messages

In the last activity, we identified potential co-conspirators by analyzing Peter's emails. We'll now continue to search for more evidence by examining Peter's SMS messages.

- SMS stands for "short message service" and is a person-to-person communication method.
- SMS messages can be no more than 160 characters.
- SMS messages can and have been used in DoS attacks.
- Smishing is the term used to refer to a social engineering attack performed using SMS messages.

Forensic Analysis of SMS Messages Demonstration

Now, we will perform a short demonstration to view the SMS entries contained in the iPhone image.

There are four methods we can use to view these entries:

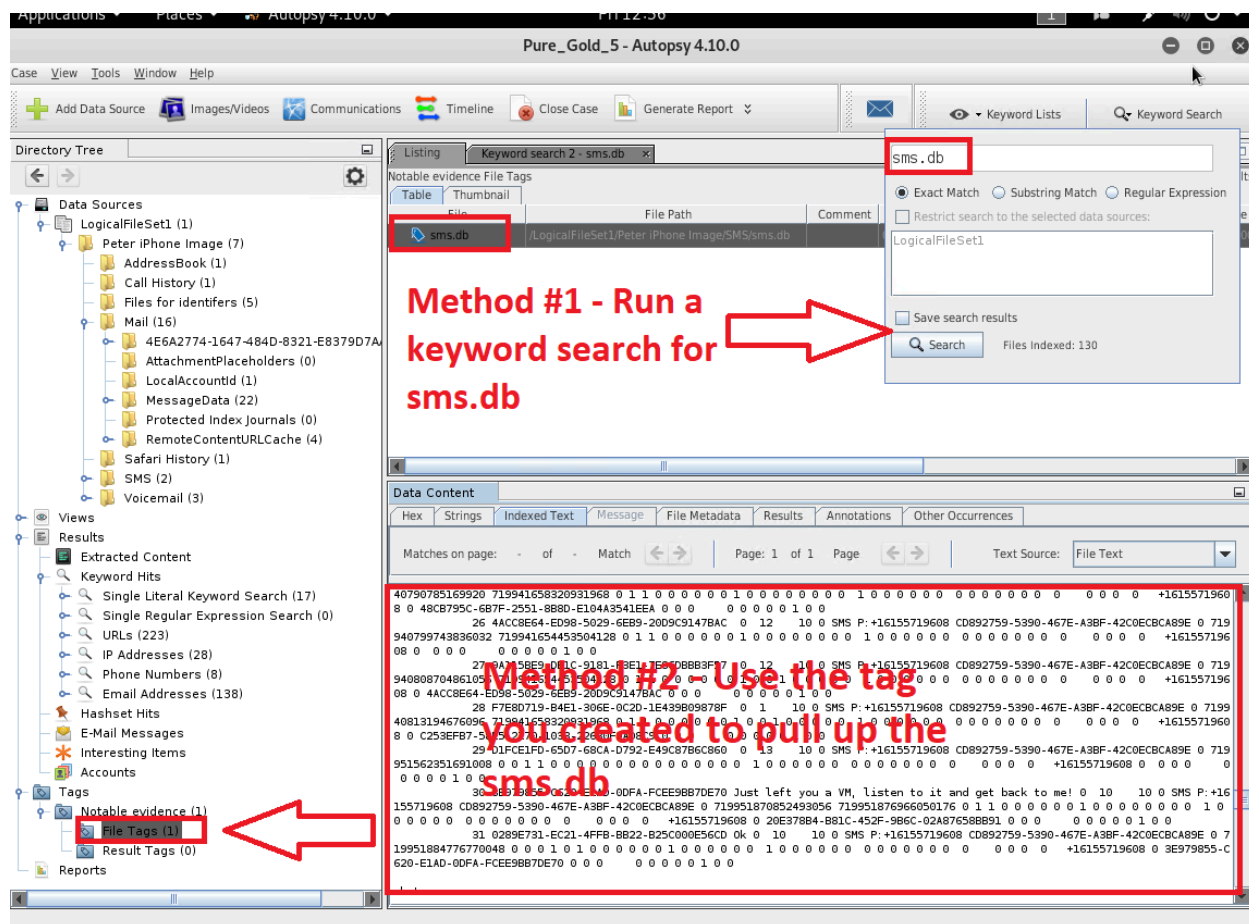
Method 1

1. From the Tools menu, select File Search by Attribute.
2. Click the box next to Name and type `sms.db`.
3. Click the Search button.
4. Select `sms.db` from the Listing pane.
5. Select the Indexed Text tab in the Data Content pane to see Peter's messages.

Method 2

Click the custom tag that you created earlier to load the `sms.db` file.

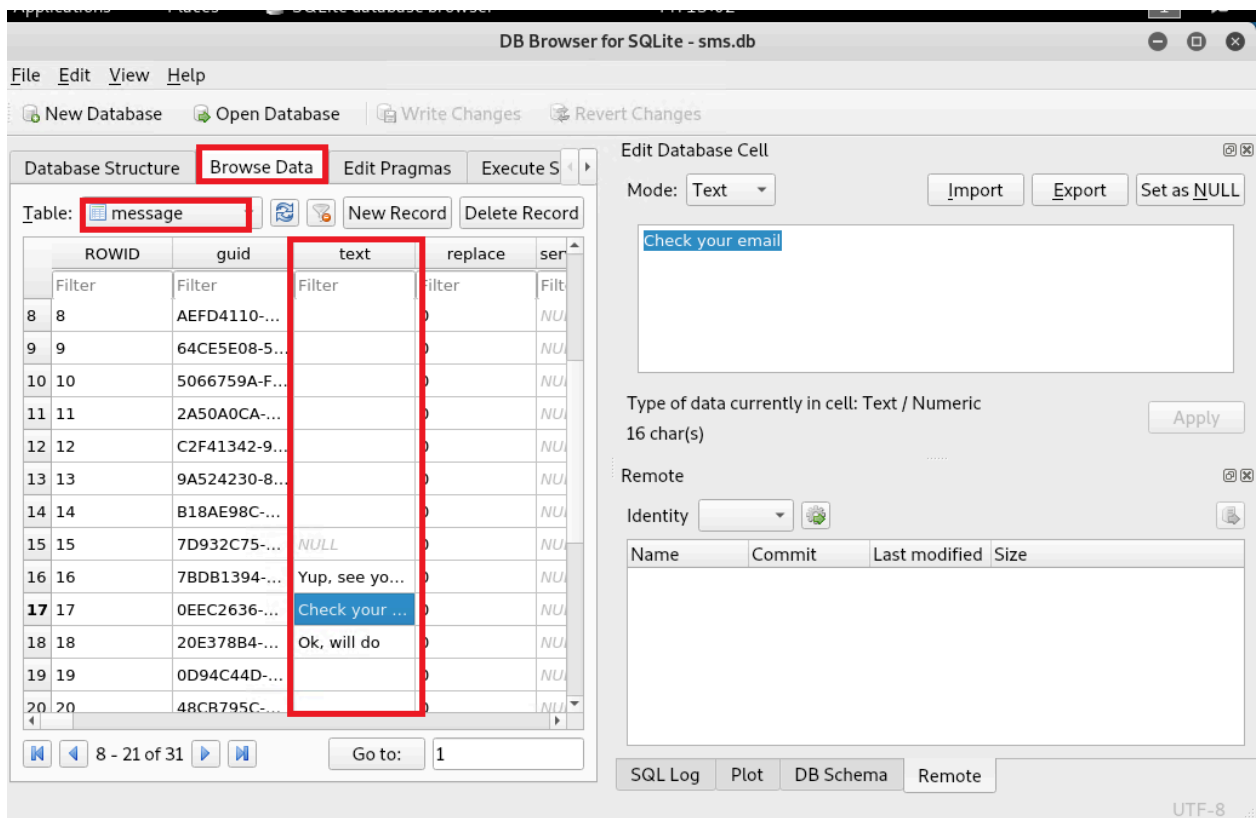
- We can read this in the Data Content pane.



Method 3

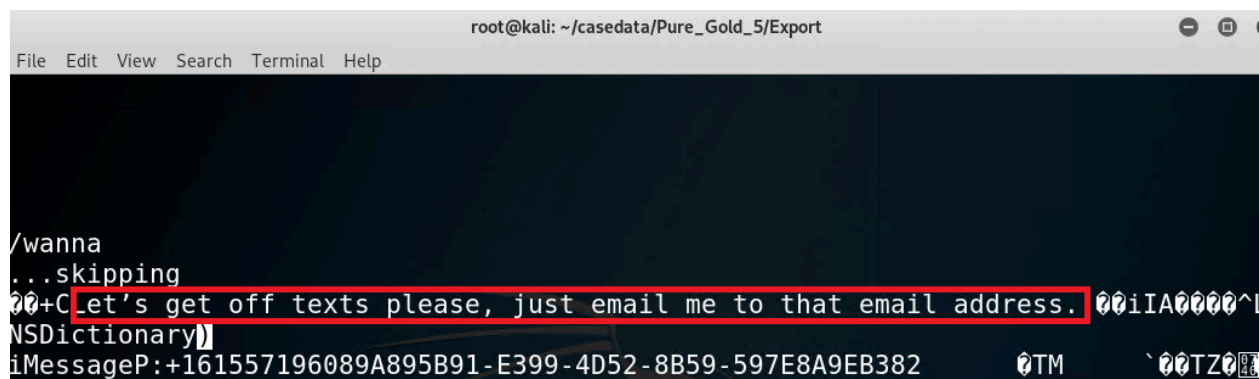
Since this is a database file, we can also use a tool like SQLite Browser to read the messages.

- Use the sqlitebrowser command to open the sms.db file
- Navigate to Browse Data and then the message table.



Method 4

Alternatively, we can export the file and use a preview command to read the messages.



- With this method, it's a little more difficult to determine the messages and senders.

09. Activity: Peter's SMS Messages

- In this activity, you will work with your group to examine Peter's SMS messages and gather more information about the case.
 - Keep your Correspondence Evidence Worksheet updated with your findings.
 - You will also export the `sms.db` file so other departments can use third-party software to perform analysis.
 - [Activity File: Peter's SMS Messages](#)
-

10. Activity Review: Peter's SMS Messages

11. Activity: The Final Report

In this activity, you will conclude your investigation by working in groups to analyze your findings and prepare a final report.

- You can continue working on whatever you don't finish at home as part of your Challenge deliverable.

Note the following:

- So far, you've examined quite a bit of information in the iPhone image and worked through the following steps:
 - Performed mobile forensic analysis and compiled details from Peter's iPhone
 - Searched through numerous files and directories on the iPhone image
 - Tagged and categorized evidence relevant to the case
 - Created custom tags
 - Extracted data for offline analysis using the export function
 - Examined and documented Peter's emails
 - Examined and documented Peter's SMS messages

Use the following resources to guide your work:

- [Worksheet: Location Information Worksheet](#)
- [Worksheet: The Final Report](#)
- [Resource: iPhone Forensic Image - Important Files and Databases](#)
- [Activity File: Final Report](#)

12. Final Report

13. Class Wrap-Up

You will continue to work on Peter's iPhone case and produce a final report as a deliverable.

Note the following key takeaways from this lesson:

- Digital forensics is dedicated to identifying, extracting, preserving, and reporting information obtained from computer and network systems.
- It relies on examiner expertise to analyze and interpret data retrieved by trusted forensics examination tools.
- Investigative teams may be spread across several time zones, so it's important to follow a baseline time zone as indicated by the case file.